NOTE

The Stuxnet computer virus shows how dangerous it can be when industrial control systems are compromised. Considered the worlds first digital weapon, this virus targeted Siemens S7-417 PLCs in SCADA systems and was reportedly responsible for the destruction of centrifuges within Iranian nuclear enrichment facilities in 2009.

OT environments use diverse proprietary protocols that evolved independently across industrial domains. A single facility might use Modbus, Profibus, Ethernet/IP, and OPC-UA simultaneously. Many legacy systems only support polling-based retrieval or require complete dataset downloads rather than selective queries, forcing digital twins to import massive amounts of data simply because source systems cannot filter at the protocol level.

Direct querying of OT systems also risks operational continuity. Industrial control systems prioritize deterministic responses over analytical workloads, and even lightweight queries can introduce dangerous latency. Process historians often lack modern query optimization, causing performance degradation when supporting both real-time ingestion and analytical queries.

These constraints typically force digital twin architectures to implement data mirroring strategies, where collectors replicate OT data to separate analytical environments. While this introduces latency and storage overhead, it protects operational systems while providing digital twins access to rich industrial data.

ARCHITECTURAL PATTERNS FOR CONVERGENCE

The industrial *demilitarized zone (DMZ)* has become the standard pattern for connecting the IT systems, where digital twin applications normally live, to the OT network. Just like a DMZ in military terms, it provides a neutral zone between the trusted OT network and the hostile, untrusted IT network. This zone hosts systems that both domains can access without directly connecting including:

- *Mirror servers* - replicate OT application interfaces for IT access without touching OT systems.
- *Protocol converters* - translate between OT protocols such as Modbus and IT protocols such as MQTT.
- *Security gateways* - inspect and filter traffic, ensuring only authorized data flows between domains.

Figure 4.5 shows a common architecture for IT/OT convergence, known as the *Purdue model*, which organizes industrial systems into hierarchical levels from field devices up to enterprise systems, enabling secure and efficient data flow while maintaining operational safety and security.